

Project Report

HR Analytics

Employee Attrition Analysis Using Microsoft Excel

Prepared By-

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Technology	Purpose
Microsoft Excel 2019	Data analysis and dashboard development
Power Query	Data cleaning and transformation
Pivot Tables	Data summarization and business analysis
Pivot Charts	Interactive data visualization
Excel Formulas	KPI calculations and analytical metrics
Conditional Formatting	Highlighting key insights and trends
Slicers	Interactive dashboard filtering
Git	Version control
GitHub	Project repository and portfolio hosting

1. Executive Summary

Executive Summary

Employee attrition has a direct impact on organizational productivity, recruitment costs, and employee morale. This project analyzes employee data to identify the major factors influencing attrition and employee satisfaction using Microsoft Excel.

The analysis includes data cleaning, exploratory analysis, pivot tables, correlation analysis, scatter plots, and interactive dashboards. Twenty business questions were answered to understand workforce behavior and identify opportunities to improve employee retention.

The project concludes with actionable recommendations that HR managers can use to reduce employee turnover and improve employee engagement.

2. Project Overview

Project Overview

This HR Analytics project aims to analyze employee attrition and workforce characteristics using Microsoft Excel.

The project focuses on:

- Understanding employee attrition
- Identifying factors affecting employee retention
- Measuring employee satisfaction
- Analyzing compensation and experience
- Providing business recommendations

The analysis was performed entirely in Microsoft Excel using Pivot Tables, Pivot Charts, Conditional Formatting, Power Query, Excel formulas, and interactive dashboards.

3. Business Objectives

Business Objectives

The primary objectives of this project are:

- Calculate the overall employee attrition rate.
- Identify departments and job roles with high attrition.
- Analyze employee demographics.

- Understand factors influencing attrition.
- Study employee satisfaction and performance.
- Build interactive dashboards for HR decision-making.
- Provide actionable business recommendations.

4. Dataset Overview

Dataset Overview

Attribute	Value
Dataset	Adecco India HR Analytics Employee Attrition
Total Rows	1,470
Total Columns	35
Blank Rows	0
Duplicate Rows	0

Target Variable

Attrition

- Yes
- No

5. Data Cleaning & Preparation

Data Cleaning Steps

The following preprocessing steps were completed before analysis:

- Verified row and column counts.
- Checked missing values.
- Checked duplicate records.
- Standardized data formats.
- Created **Attrition_Flag** (Yes = 1, No = 0).

- Built calculated columns required for analysis.
- Created Pivot Tables for business questions.
- Developed interactive Pivot Charts.

The dataset was ready for analysis without major quality issues.

6. Key Business Questions

Basic Analysis

The following business questions were answered:

1. Overall attrition rate
 2. Department with highest attrition
 3. Average age of employees leaving
 4. Job satisfaction by job role
 5. Attrition by gender
 6. Average income of employees leaving
 7. Distance from home vs attrition
 8. Performance rating distribution
 9. Overtime distribution
 10. Average years at company
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Advanced Analysis

Additional analyses included:

- Correlation analysis with attrition
- Job involvement vs attrition
- Age vs job satisfaction
- Marital status vs attrition
- Training impact
- Work-life balance vs performance
- Stock option impact

- Years at company vs satisfaction
 - Business travel vs satisfaction
 - Promotion vs performance
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7. Key Findings & Insights

Key Findings

Attrition

- Overall employee attrition rate is **16.12%**.
 - Sales department experiences the highest employee attrition.
 - Single employees have significantly higher attrition than married and divorced employees.
 - Employees working overtime are more likely to leave.
 - Employees without stock options show higher attrition.
 - Employees receiving no training have the highest attrition.
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Employee Satisfaction

- Job satisfaction shows minimal relationship with employee age.
 - Years at company has little influence on satisfaction.
 - Business travel has minimal effect on job satisfaction.
 - Work-life balance has limited impact on performance ratings.
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Correlation Analysis

The strongest factors associated with attrition are:

- Total Working Years
- Job Level
- Years in Current Role
- Monthly Income
- Age

These variables show the highest correlation with employee attrition among the analyzed numerical features.

8. Recommendations

Business Recommendations

Based on the analysis, the following recommendations are suggested:

- Improve employee engagement programs.
 - Enhance career development opportunities.
 - Increase employee training initiatives.
 - Review overtime policies.
 - Expand stock option benefits where applicable.
 - Focus retention efforts on single employees and high-risk departments.
 - Conduct regular employee satisfaction surveys.
 - Monitor attrition trends using interactive HR dashboards.
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10. Conclusion

Conclusion

This project demonstrates how Microsoft Excel can be used to perform comprehensive HR Analytics without specialized analytics software.

Through data cleaning, exploratory analysis, pivot tables, correlation analysis, and interactive dashboards, the project successfully identified key factors influencing employee attrition and employee satisfaction.

The insights generated can help HR managers make informed decisions regarding employee engagement, retention strategies, workforce planning, and organizational performance.

Overall, the project provides a practical example of using Excel for business intelligence and HR decision support.